

产品定义 Product Definition

E-Series Electrical Test Connectors offer long-life, quick connections between the pallet C9M docking module and customer interfaces. Suitable for low-voltage signals, drive currents, and high-current probes, with customizable wiring.

E10 Series: User-friendly, manual connection.

E20 Series: Bottom-pallet or cylinder mounted for auto/semi-auto connection.

E10T: Signal

E10D / E10DZ: Circular pin / Circular socket

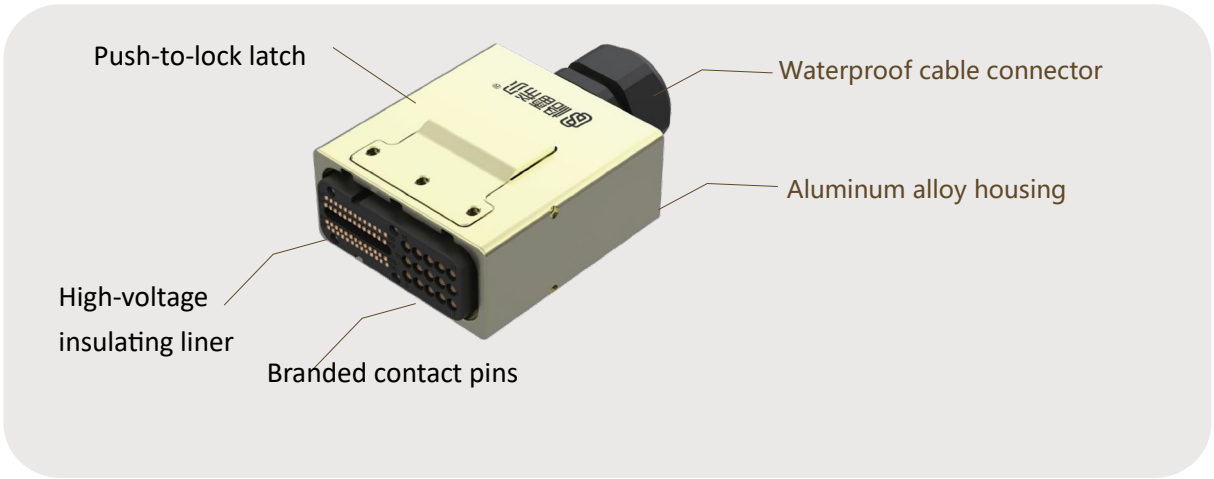
E10JP / E10J: Flat electrode / Clip-on female

E10U: UVW copper busbars

Custom: Copper lugs, high-current probes, and high-voltage solutions available.

Core Advantage: High Mating Cycle Life Unlike traditional connectors limited to 50-500 cycles, the E10 is designed for high mating cycle life. It ensures stable electrical performance under high-frequency use, significantly reducing failure rates and securing smooth production.

E10 电连接器结构示意 Structural Diagram of E10 Electrical Connector



E10T Signal Connector E10T for Signal Connector

The E10T signal connector features an extensive product portfolio, with most based on customized interfaces. Equipped with suitable probes and structures, it delivers a service life of over 100,000 cycles.

Once the interface is finalized, customers may have specific requirements regarding connector style (straight-through / 90° elbow), shielding, high-voltage capability, wire type, wire length and other parameters. Consequently, the actual product models vary greatly. In-depth technical discussion and detailed confirmation with engineers are therefore required.

E10T Series Model Code Explanation：

E10T-SSSS-XXP(H)-Lyyy

SSSS: Project Number

XX: Number of Pins

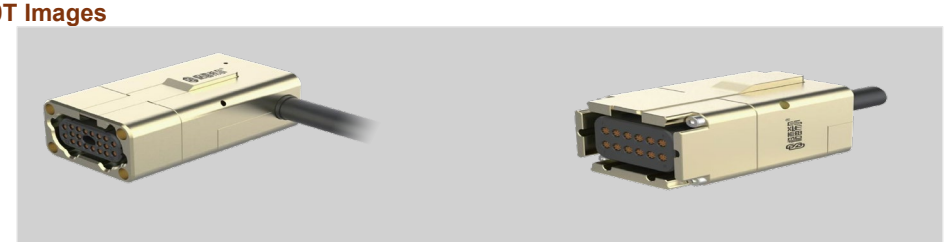
H: Mixed type with pins of different sizes

yyy: Wire Length

Technical Specification Options

Specification Item	Default	Optional	Remarks
1 Probe Brand	Domestic	Imported	/
2 Wire Type	300-600V tinned high-temperature wire	3KV high-temperature silicone flexible wire	Wire gauge depends on interface and customer requirements
3 Ground Shielding	Non-shielded	Shielded	/
4 Wire Configuration	Single wire combination / single wire twisted pair combination	Cable (single type wire only)	Mixed wire types are not applicable for cables
5 Wire Outlet Direction	Straight through	90° elbow	Elbow direction to be specified
6 Mounting Method	E10T Manual	E20T Tray or Cylinder Mounting	/
7 Wire Termination	Numbered bare wire	Aviation plug	Other heavy-duty connectors available
8 Sensor	None	Pin-in-place switch sensor	/

E10T Images



E10D / E10DZ High-Voltage Connector

The E10D connector is designed for high-current circular pin interfaces. The E10D side uses a long-life socket contact. For different connector types with pin diameters of 8 mm, 10 mm, 12 mm, and 14 mm, corresponding high-cycle test solutions are available.

Compared with standard automotive connectors, the E10D series uses high-reliability socket contacts with gold-plated or silver-plated surfaces. The typical mating life exceeds 100,000 insertion cycles.

The E10DZ connector is designed for circular socket interfaces. Unlike the E10D, the E10DZ side uses long-life pin contacts.

Pin diameters are available from 8 mm to 14 mm. The long service life is achieved through wear-resistant coatings and high-conductivity materials, providing excellent reliability and significantly longer mating life than standard connectors.

E10D Part Number Description

E10D-SSSS-XXPDDD-LYYY (For Pin Interface Connection)

E10DZ-SSSS-XXPDDD-LYYY (For Socket Interface Connection)

SSSS Project Number

XX Number of Contacts (typically 2 or 3 contacts with HVIL signal)

DDD Pin Size Code

LYYY Cable Length



Technical Specifications

Specification Item	Default	Optional	Remarks
1 Cable Type	EV Single Cable (Non-shielded)	EVP Single Cable (Shielded)	1500V Voltage Resistance
2 Insulation Voltage Test	No	10KV Voltage Resistant Cable	
3 HVIL (High Voltage Interlock)	Lead out	No lead out	
4 Cable Outlet Direction	Straight through	90-degree elbow	Direction must be specified for elbows
5 Installation Method	E10D/E10DZ Manual	E20D/E20DZ Tray or Cylinder Installation	
6 Wire Terminal	Bare wire	Copper lug	Other types customizable
		High-current quick plug	
		High-current quick plug	
		Voltage-resistant cross plug	
		Voltage-resistant triangle plug	
7 Sensor	None	Temperature sensor	
		Switch pin in-position sensor	

E10D/E10DZ for High Voltage Connector

The E10JP electrical test connector is designed for tab-type high-current interfaces, fitted with clip pins or tab electrodes on the E10JP side. Compared with original automotive connectors, it is well suited for testing scenarios requiring frequent operation, long service life and high current load. For medium current ranging from 40A to 80A, it adopts a clip probe mechanism. For heavy current of 150A to 400A, it uses the Gripseal self-adaptive clip structure, which delivers a far longer service life.

The E10J electrical test connector is developed for clip female interfaces, with tab electrodes on the E10J side. These electrodes are made of high-strength beryllium copper with surface plating treatment to enhance wear resistance and achieve long service life.

E10JP Model Coding Explanation

E10JP-SSSS-XXPDDD-Lyyy (for tab electrode connection)

E10J-SSSS-XXPDDD-Lyyy (for clip female connector connection)

SSSS: Project number

XX: Number of pins

DDD: Rated current

Yyy: Cable length

Specification Options:

Specification Item	Default Configuration	Optional Configuration	Remarks
1. Cable Type	EV Single Cable (Unshielded)	EVP Single Cable （Shielded）	1500V withstand voltage
2. Insulation Withstand Voltage Test	No	10kV Withstand Voltage Cable	Small wire gauge
3. High-Voltage Interlock	Wired out	Not wired out	—
4. Cable Outlet Direction	Straight outlet	90°elbow outlet	Elbow direction must be specified
5. Mounting Method	Manual E10JP Type	E20JP Tray/Cylinder Mounted Type	—
6. Cable Termination	Bare wire	Cable lug Large-current Quick Connector Cross-type Triangular Withstand Voltage	Other types can be customized Insulation withstand voltage test Insulation withstand voltage test
7. Customer Electrical Interface Type	E10JP for tab electrode connection	E10J for clip female connector connection	—
8. Sensor	None	Temperature sensor Pin-in-place detection sensor	—





E10 for UVW Copper Busbars

E10U busbar interface electrical test connector is our high durability, fast electrical connector designed for busbars on the market. It typically designed in 3-pin or 6-pin configurations, and is also available in other types like Type 2. It uses a handle-rotation connection (no screw required), ensuring reliable connection and easy operation. It’ s commonly used in high-current applications.

Encoding Description of E10U

E10U-SSSS-XXP-Xddd-LyyySSSS

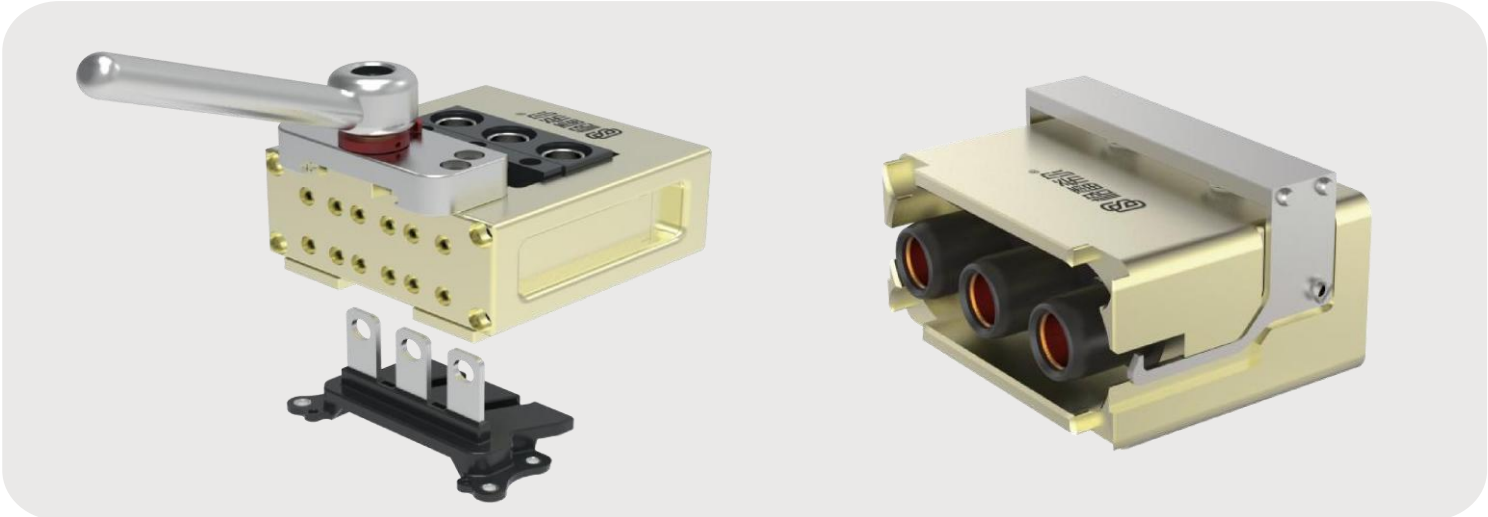
SSSS project number

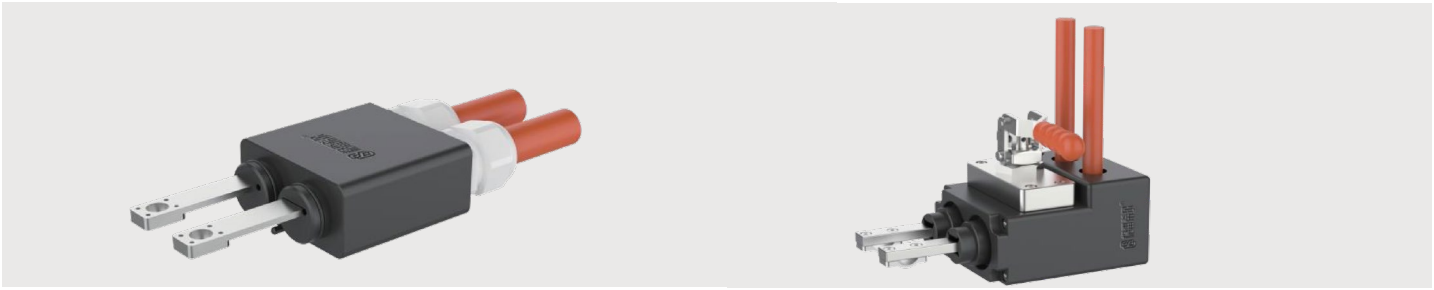
XX Quantity of Pins (Typically 3 or 6)

Xddd Wire Size Code

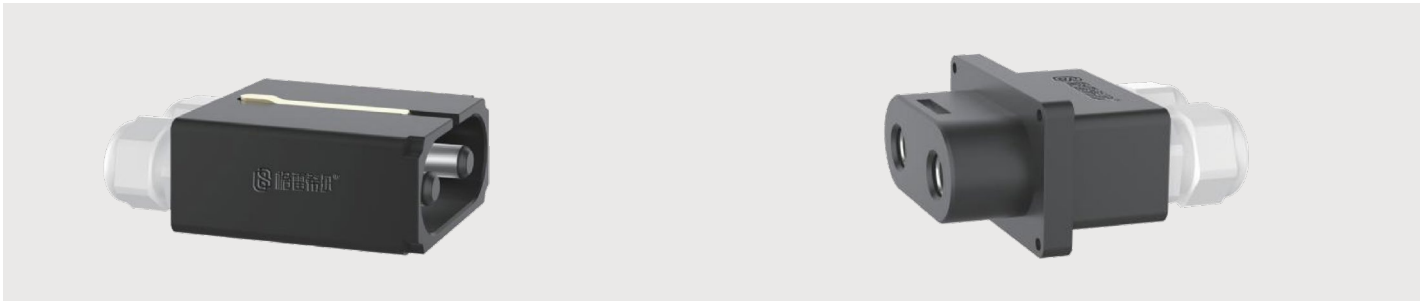
Yyy Wire Length

Specification Item	Default Configuration	Optional Configuration	Remarks
Cable Type	EV Single Cable (Unshielded)	EVP Single Cable (Shielded)	1500V Withstand
Insulation Withstand Voltage	No	10KV Withstand Voltage Cable	Small Wire Diameter
High-Voltage Interlock	Lead-out	Non Lead-out	-
Cable Outlet Direction	Straight-through	90° Elbow	Elbow direction fixed
Installation Method	E10U Manual	E20U Pallet or Cylinder Mount	-
6Wire Termination	Bare Wire	Copper Lug	Other types





E10 terminal quick connector



E10C High-Amperage Male-Female Terminal



E-series Connector for Insulation and Dielectric Withstand Voltage Test

The E-series electrical test connectors are a crucial part of the C9 docking plate assembly. They offer fast, long - lasting, convenient and reliable electrical connections, and their application scenarios are vastly different from those of ordinary automotive connectors. The advantages include:

- ●Long - lasting insertion and extraction performance
- ●Maintainability
- ●One – second connection and instantaneous testing
- ●Low operation cost

E-series electrical test connectors are customizable. We' ll need to discuss these details:

- ●3D drawings of the electrical interface
- ●Interfaces to connect (quantities)
- ●Technical specification options
- ●Wire length
- ●Other special requirements

After order confirmation, please also provide:

- ●Final 3D drawings/physical samples of the electrical interface
- ●Additional requirements (e.g., testing)
- ●Confirmation on whether to make samples first